

Designing the Referral Table: Attribution & Channel Taxonomy

How a 422-referrer traffic log became a 6-channel lookup table Mike can use to set the Q4 ad budget

Context

Breaking Games sells direct-to-consumer through breakinggames.com. Between January and July 2025, the company spent \$37,200 on Meta (Facebook and Instagram) ads and recorded \$61,085 in tracked sales. Site traffic over that window arrives from 422 distinct referrer strings, logged in a single table (referrals) with no notion of “channel” built in — just a raw referrer value and a landing page for every visit summary.

Of those 422 referrers, the top 21 carry 99% of all pageviews; the rest is long-tail noise — forum threads, blog links, and one-off sites with a handful of visits each. Nothing in the raw data says which of the 422 belong together, and two traps sit directly in the way of a naive grouping: names that look related but aren't (google is free organic search; googlesyndication is Google's paid display network), and names that look unrelated but should be grouped (facebook, instagram, and meta are three doors into the one company Breaking Games actually pays for advertising).

Business Problem

Mike, Breaking Games' Head of E-Commerce, has a Q4 budget review with Shari and one question he can't currently answer: did the Meta ad spend work, and where should the next dollar go — the same campaigns at higher volume, a different set of campaigns, or an entirely new channel? He can't answer it because ad spend (in `meta_campaigns`) and site traffic (in `referrals`) live in disconnected systems with no shared key, and the traffic side isn't even organized into channels yet. Before any spend-versus-traffic comparison is possible, the 422 raw referrer values have to become a small set of clean, decision-relevant categories.

My Approach

1. Establish what the data can and can't support. Before designing any categories, I checked whether Breaking Games' data could run a formal attribution model (last-touch, first-touch, linear, time-decay, position-based). It can't: there are no UTM tags, no session-to-purchase link, and each row in `referrals` summarizes a referrer/landing-page pair, not one visitor's path across a journey. I treated this as a hard constraint on the whole project, not a detail to smooth over — what the data supports is correlation (“this much Facebook traffic happened alongside this much revenue”), not causal attribution, and every downstream number needed to carry that caveat honestly.
2. Work from the real data, not assumptions. I pulled the top 50 referrers ranked by pageviews and built the taxonomy from what was actually there, rather than naming categories in the abstract and hoping the data fit them.
3. Draft categories, then pressure-test each one against volume. My first draft included a dedicated “Community” category for hobbyist and tabletop-specific sites (BoardGameGeek, convention platforms,

hobby blogs). Querying the real numbers showed it carried 2,914 pageviews — 0.87% of all traffic — and converted at roughly the site average, with more than half of that volume sitting in a single referrer. It didn't clear the bar of being worth telling apart, so I cut it and folded those eleven referrers into the catch-all rather than keep a category that looked meaningful but wasn't.

4. Validate before writing anything. I computed the projected pageview share for every proposed channel against the full referrals table before writing a single UPDATE statement — specifically checking that the catch-all (“Other”) stayed under the few-percent threshold that would signal a real channel had been missed.
5. Hold execution back until the taxonomy is confirmed. I wrote the `dim_referrer` build script but did not run it against the live database. It's handed off for review and manual execution, so nothing is written until the taxonomy itself has been explicitly signed off on.

Solution

A lookup table, `dim_referrer`, with three columns — `referrer` (the raw string exactly as it appears in the data), `channel` (the clean bucket), and `channel_type` (Paid, Organic, Owned, or Other) — built from a six-channel taxonomy validated against 333,550 total pageviews:

Channel	Includes	Share of pageviews	Why this split
Owned	direct, klaviyo, gmail, live, office, deref-gmx, yahoo! mail, outlook.com, inbox, mailchi	46.7%	Breaking Games' own audience returning on its own — the baseline no ad channel should get credit for.
Paid Social	facebook, instagram, meta	27.3%	The only channel with matched spend data (<code>meta_campaigns</code> / <code>meta_ad_sets</code>). Every Q4 dollar decision starts here.
Organic Search	google, bing, duckduckgo, yahoo!, brave, ecosia, yandex, baidu, qwant, naver, mojeek, chatgpt, perplexity	24.2%	Free demand Breaking Games didn't pay for — the baseline for judging whether paid search is worth testing.
Other	388 remaining referrers — long tail plus likely bot/crawler traffic	1.4%	Catch-all, verified small enough that no real channel is hiding inside it.
Organic Social	tiktok, pinterest, reddit, linkedin, nextdoor, youtube	0.3%	Unpaid traffic on platforms Breaking Games isn't currently buying ads on — the evidence base for “a whole new channel.”
Paid Search	googlesyndication, syndicatedsearch	0.3%	Non-Meta paid placements, kept separate so they're never folded into “google” and miscounted as free.

Key decisions and why

Decision point	Choice made	Reasoning
Community category (BGG, conventions, hobby sites)	Cut; folded into Other	0.87% of pageviews, average conversion rate, 52% concentrated in one referrer — didn't meet the “worth telling apart” bar.
google vs. googlesyndication	Kept as separate channels (Organic Search vs. Paid Search)	Same word, opposite economics — one is free, one is paid. Merging them would mix spend-driven and free traffic in the same bucket Mike is making a budget call on.
facebook / instagram / meta	Merged into one channel (Paid Social)	Three surfaces of the one platform Breaking Games actually spends ad dollars on — splitting them would fragment the only channel with real spend data behind it.
youtube	Organic Social, not Organic Search	Functions as a content/video platform rather than a search engine; large enough (577 pageviews) that misclassifying it would move a real number.
nytimes (single earned-media hit)	Left in Other, not split into its own “Earned Media” category	N of 1 — not enough evidence yet to justify a dedicated bucket. Revisit if press mentions become a recurring source.

Business Insight / Impact

- Owned traffic (46.7%) is nearly double Paid Social (27.3%). Breaking Games' brand recognition and email list are doing more work than its ad spend — a fact that should temper any instinct to solve the Q4 question by simply increasing Meta spend.
- Paid Social and Organic Search are close in size (27.3% vs. 24.2%). Free search demand is nearly as large as the paid channel Mike is reviewing — worth knowing before assuming ads are the dominant driver of site traffic.
- Non-Meta paid and social channels combined are under 1% of traffic (Paid Search 0.3% + Organic Social 0.3%). That's either a genuinely untapped opportunity or evidence Breaking Games' audience simply isn't there yet — the next step in this project (the Q4 allocation plan) needs to interrogate which, not assume it.
- What this table enables: a single GROUP BY channel query against referrals now answers “what share of traffic and checkouts does each real channel drive,” replacing 422 unreadable referrer names with six categories Mike can act on. This is the direct input to the CTR/CPC/ROI campaign ranking and Q4 allocation plan that come next in this project.
- What this table doesn't solve yet: it describes channel volume, not ad efficiency. Joined against meta_campaigns spend, it still only supports a correlation-level “cost per pageview,” not true attribution or ROI — that limit has to travel with every number pulled from this table into Mike's Q4 review, stated plainly rather than implied away.